Um único estado final para AFN e AFD

Observação

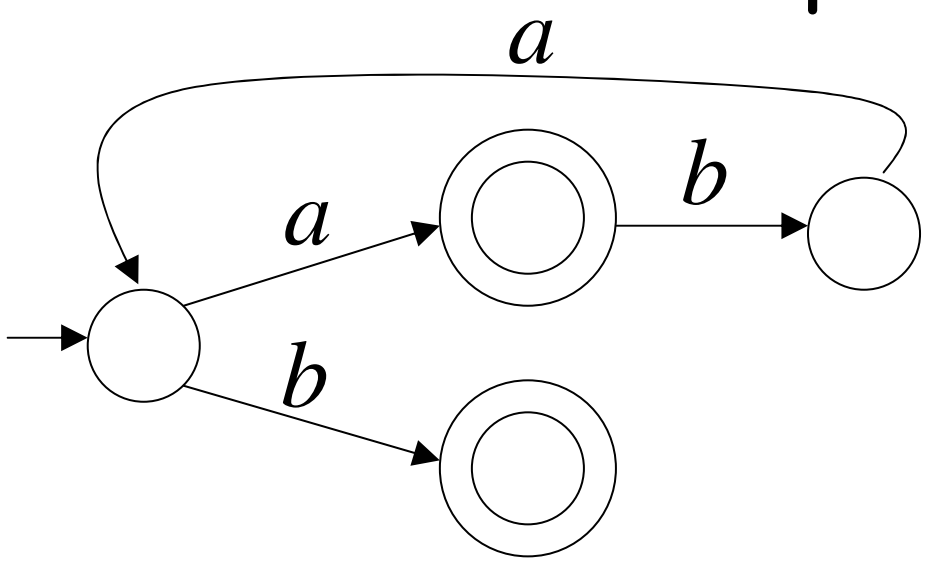
Qualquer autômato finito (AFN ou AFD)

pode ser convertido para um AFN equivalente

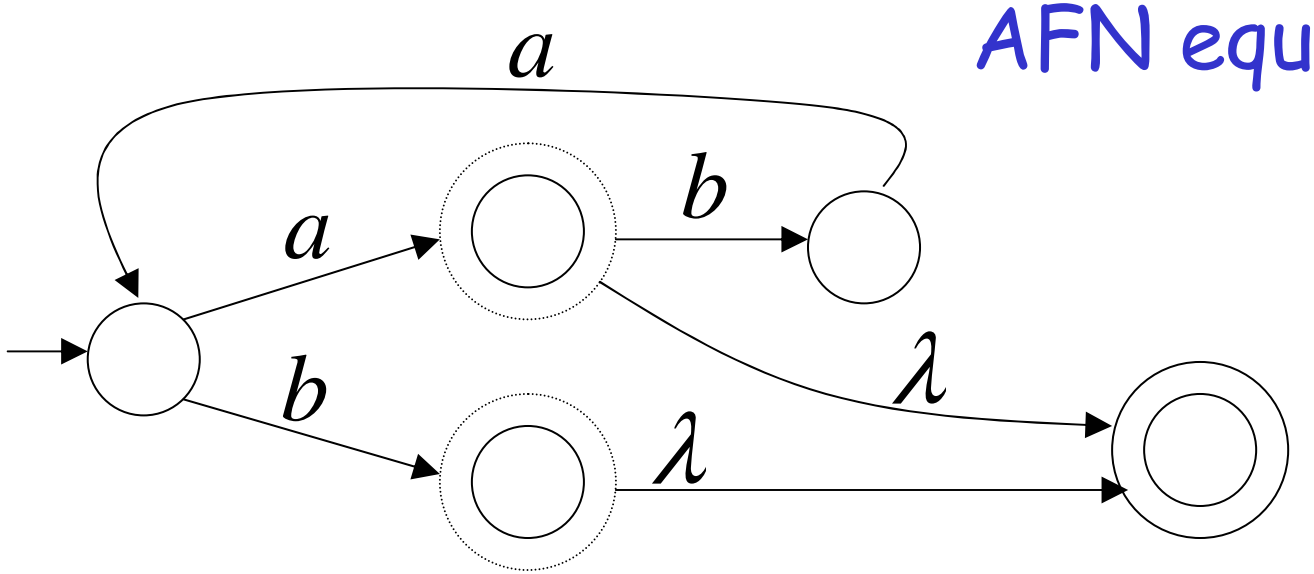
com um único estado final

Exemplo

AFN

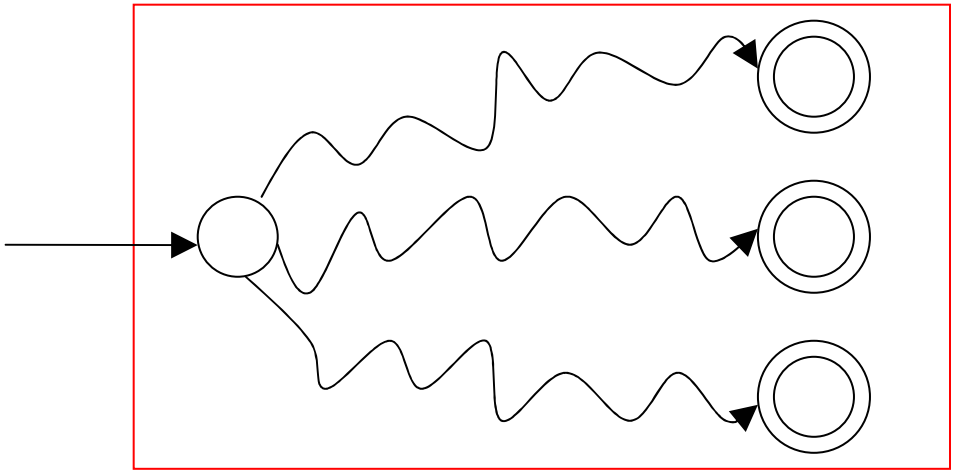


AFN equivalente



Em geral

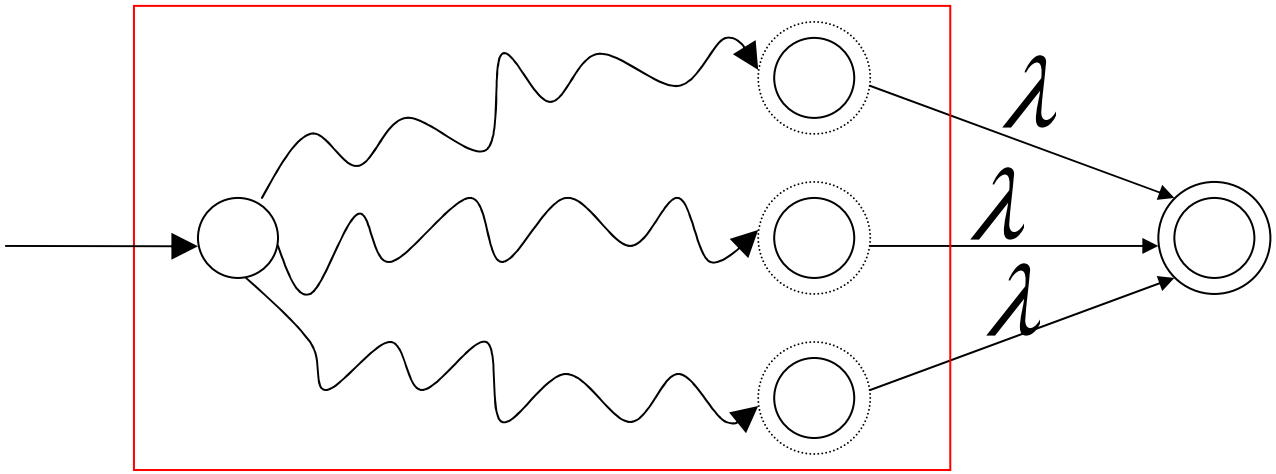
AFN



Atenção:

$\lambda = \varepsilon$

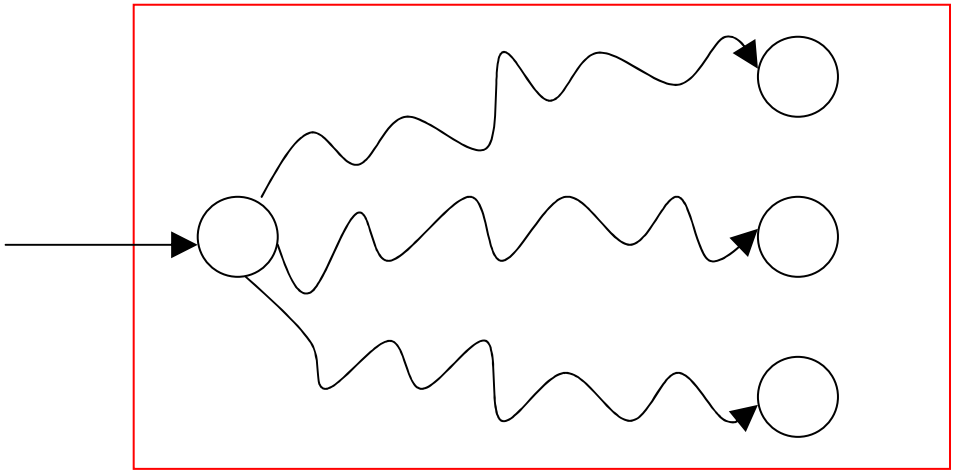
AFN equivalente

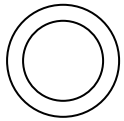


Único estado final

Caso extremo

AFN sem estado final





Adicione um estado final sem transição

Algumas propriedades de linguagens regulares

Propriedades

Para linguagens regulares L_1 e L_2 provaremos:

$$\left.\begin{array}{ll} \text{União:} & L_1 \cup L_2 \\ \\ \text{Concatenação:} & L_1 L_2 \\ \\ \text{Estrela:} & L_1{}^* \end{array}\right\} \text{São linguagens regulares}$$

Dizemos que:

Linguagens regulares são **fechadas sobre**

União: $L_1 \cup L_2$

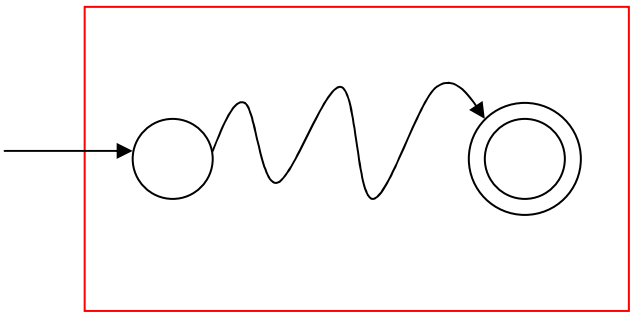
Concatenação: $L_1 L_2$

Estrela: $L_1 *$

Linguagem regular L_1

$$L(M_1) = L_1$$

AFN M_1

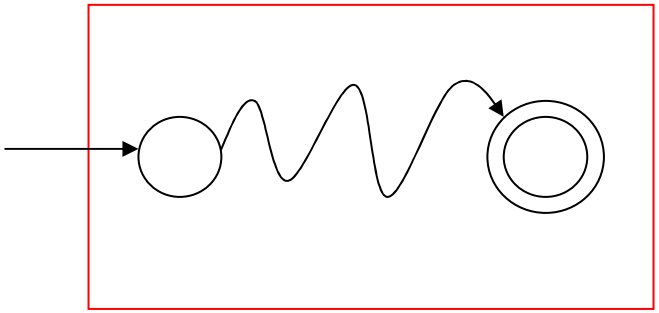


Único estado final

Linguagem regular L_2

$$L(M_2) = L_2$$

AFN M_2

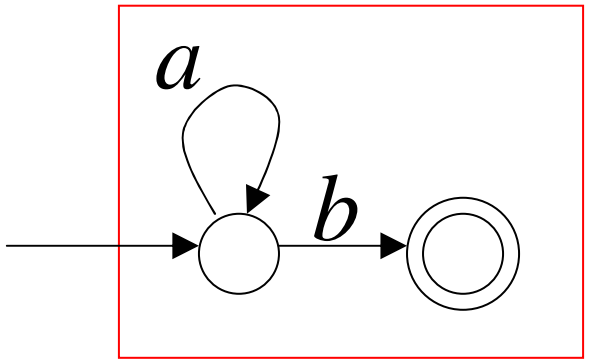


Único estado final

Exemplo

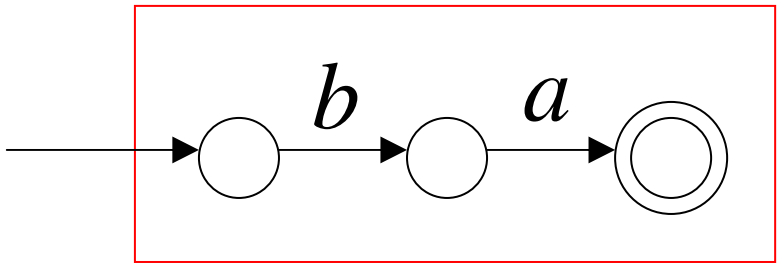
M_1

$L_1 = \{a^n b\}$



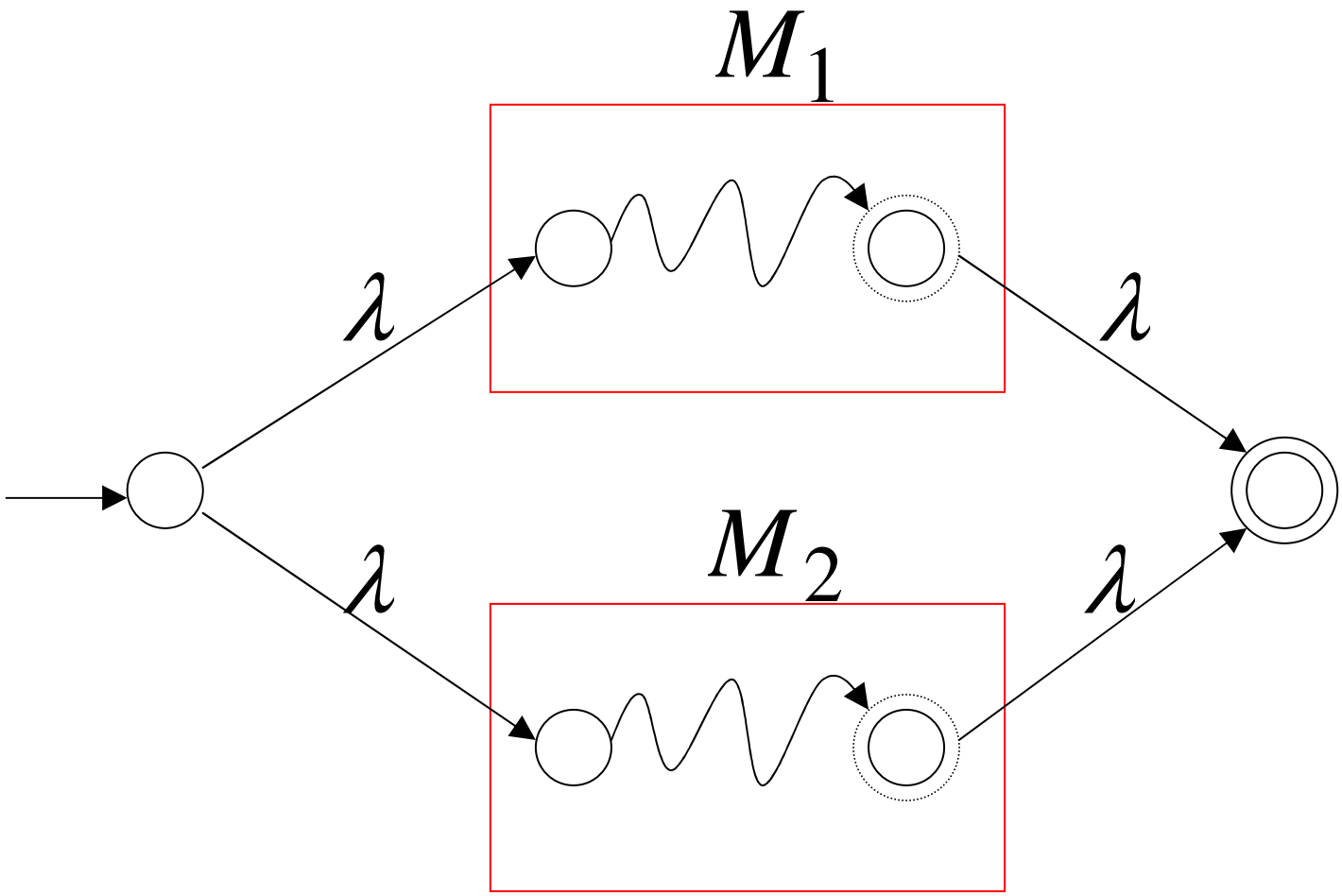
M_2

$L_2 = \{ba\}$



União

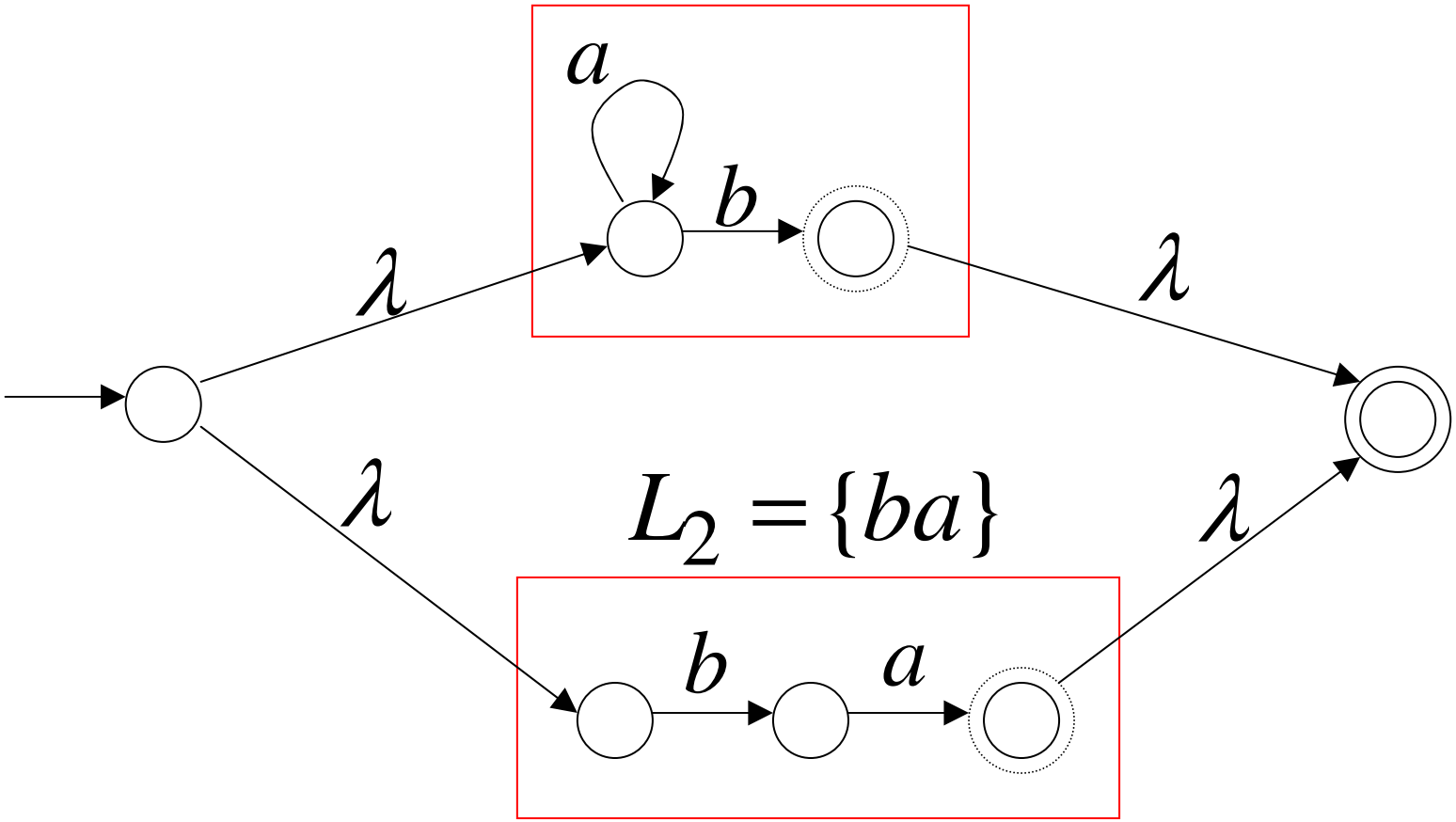
AFN para $L_1 \cup L_2$



Exemplo

AFN para $L_1 \cup L_2 = \{a^n b\} \cup \{ba\}$

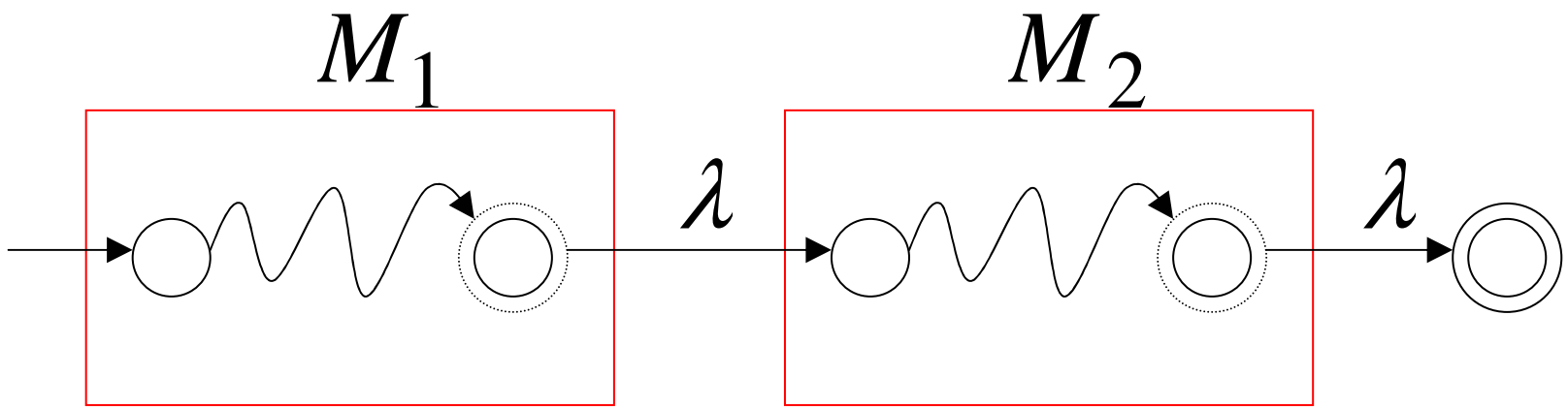
$$L_1 = \{a^n b\}$$



$$L_2 = \{ba\}$$

Concatenação

AFN para $L_1 L_2$

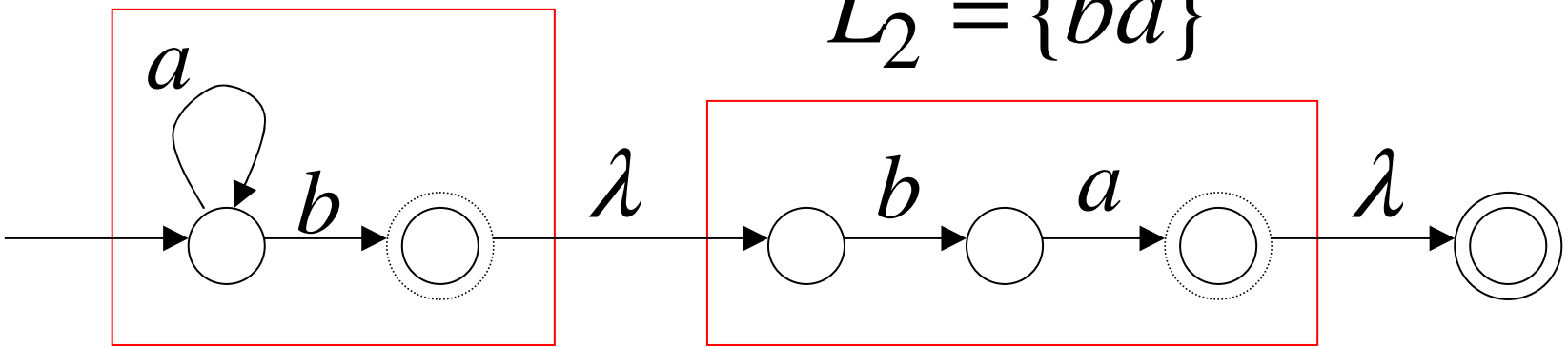


Exemplo

AFN para $L_1 L_2 = \{a^n b\}\{ba\} = \{a^n bba\}$

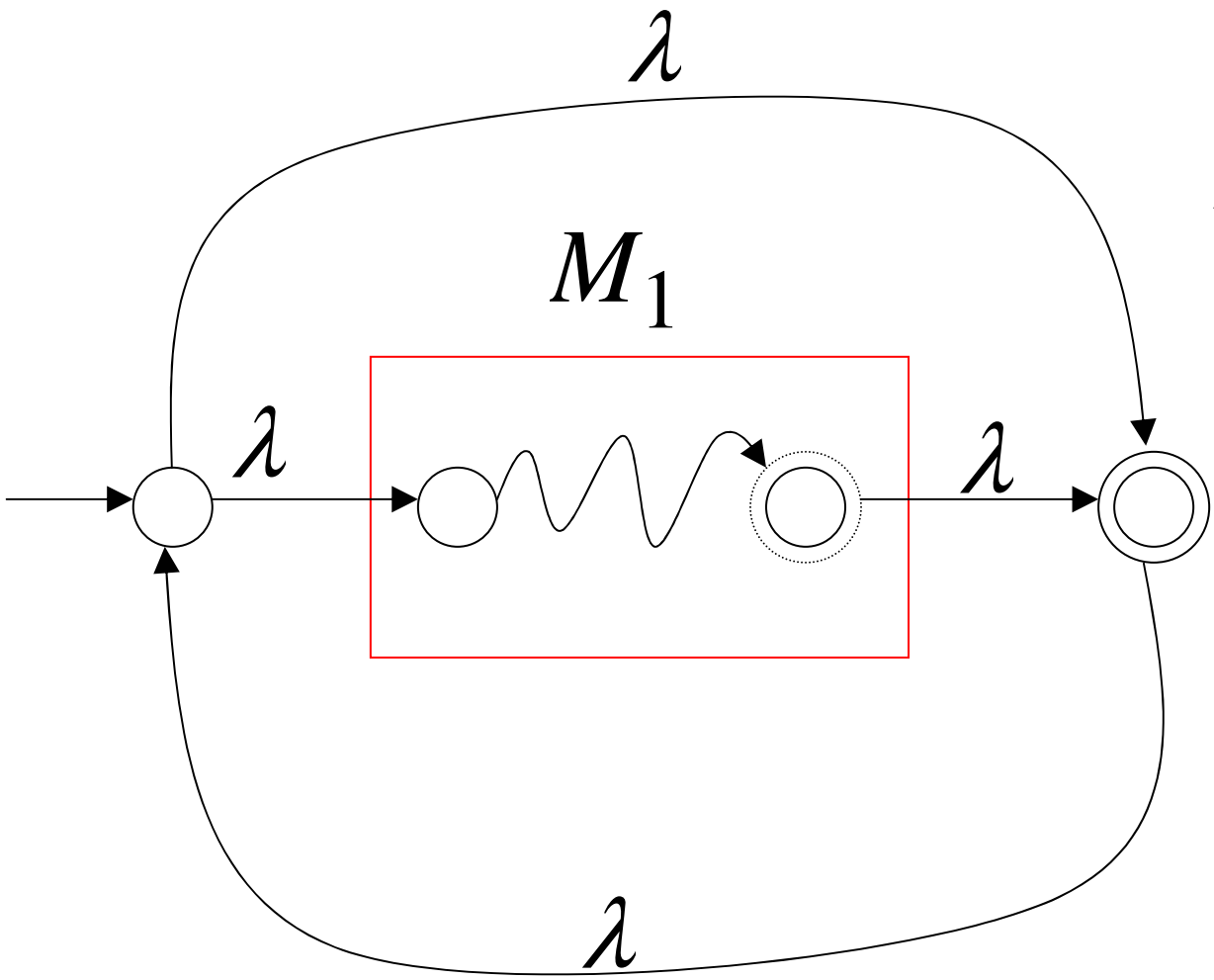
$L_1 = \{a^n b\}$

$L_2 = \{ba\}$



Operação estrela

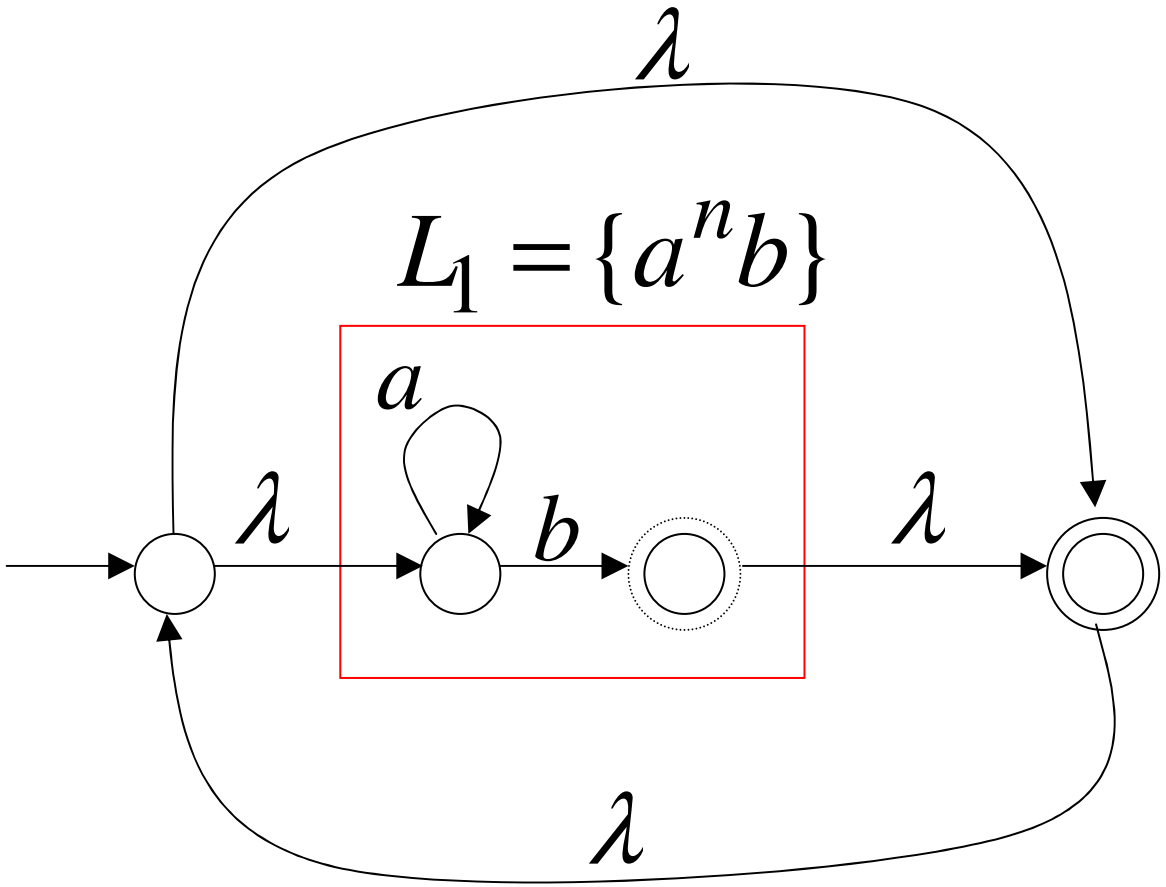
AFN para $L_1{}^*$



$\lambda \in L_1{}^*$

Exemplo

AFN para $L_1{}^* = \{a^n b\}^*$



Expressões Regulares

Expressões Regulares

Expressões regulares (ER)
descrevem linguagens regulares (LR)

Exemplo: $(a + b \cdot c)^*$

descreve a linguagem

$$\{a, bc\}^* = \{\lambda, a, bc, aa, abc, bca, \ldots\}$$

Definição Recursiva

ERs primitivas: $\varnothing, \quad \lambda, \quad \alpha$

Dada uma ER r_1 e r_2

$$\left.\begin{array}{l} r_1 + r_2 \\ r_1 \cdot r_2 \\ r_1{}^* \\ (r_1) \end{array}\right\} \text{São ERs}$$

Exemplos

É ER: $(a+b\cdot c)^*\cdot(c+\varnothing)$

Não é ER: $(a+b+)$

Linguagens de ER

$L(r)$: linguagens da ER r

Exemplo

$$L((a+b\cdot c)^*)=\{\lambda,a,bc,aa,abc,bca,\ldots\}$$

Definição

Para ERs primitivas:

$$L(\varnothing) = \varnothing$$

$$L(\lambda) = \{\lambda\}$$

$$L(a) = \{a\}$$

Definição (continuação)

Para ERs r_1 e r_2

$$L(r_1 + r_2) = L(r_1) \cup L(r_2)$$

$$L(r_1 \cdot r_2) = L(r_1)\, L(r_2)$$

$$L(r_1 *) = (L(r_1))*$$

$$L((r_1)) = L(r_1)$$

Exemplo

ER: $(a+b)\cdot a^*$

$$
\begin{aligned}
L((a+b)\cdot a^*) &= L((a+b))\,L(a^*) \\
&= L(a+b)\,L(a^*) \\
&= (L(a)\cup L(b))\,(L(a))^* \\
&= (\{a\}\cup\{b\})\,(\{a\})^* \\
&= \{a,b\}\{\lambda,a,aa,aaa,\ldots\} \\
&= \{a,aa,aaa,\ldots,b,ba,baa,\ldots\}
\end{aligned}
$$

Exemplo

ER $$r = (a+b)^*(a+bb)$$

$$L(r) = \{a, bb, aa, abb, ba, bbb, \ldots\}$$

Exemplo

ER $r = (aa)^*(bb)^*b$

$$L(r) = \{a^{2n}b^{2m}b: \quad n,m \geq 0\}$$

Exemplo

ER $r = (0+1)^* 00 (0+1)^*$

$L(r)$ = { Todas as strings com pelo menos dois 0 consecutivos }

Exemplo

ER $r = (1+01)^*(0+\lambda)$

$L(r)$ = { Todas as strings sem dois 0 consecutivos }

LR equivalentes

Definição:

As ERs r_1 e r_2

são **equivalentes** se $L(r_1) = L(r_2)$

Exemplo

L = { Todas as strings sem dois 0 consecutivos }

$$r_1 = (1+01)^*(0+\lambda)$$

$$r_2 = (1^*011^*)^*(0+\lambda)+1^*(0+\lambda)$$

$$L(r_1) = L(r_2) = L$$

r_1 e r_2 São ERs equivalentes

Expressões regulares
Linguagens regulares

Teorema

$$\left\{ \begin{array}{l} \text{Linguagens} \\ \text{Geradas por} \\ \text{Expressões Regulares} \end{array} \right\} = \left\{ \begin{array}{l} \text{Linguagens} \\ \text{Regulares} \end{array} \right\}$$

Teorema - Parte 1

$$\left\{ \begin{array}{l} \text{Linguagens} \\ \text{Geradas por} \\ \text{Expressões Regulares} \end{array} \right\} \subseteq \left\{ \begin{array}{l} \text{Linguagens} \\ \text{Regulares} \end{array} \right\}$$

1. Para qualquer expressão regular r a linguagem $L(r)$ é regular

Teorema - Parte 2

$$\left\{\begin{array}{l}\text{Linguagens}\\ \text{Geradas por}\\ \text{Expressões Regulares}\end{array}\right\} \supseteq \left\{\begin{array}{l}\text{Linguagens}\\ \text{Regulares}\end{array}\right\}$$

2. Para qualquer linguagem L existe uma ER r com $L(r) = L$

Prova - Parte 1

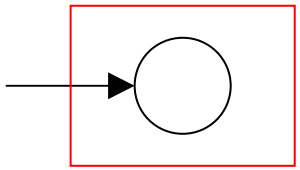
1. Para qualquer ER r
   a linguagem $L(r)$ é regular

Prova por indução no tamanho de r

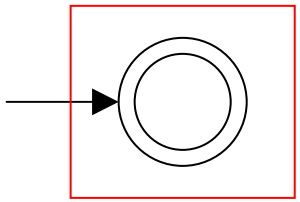
Indutiva Básica

ERs primitivas: $\varnothing, \quad \lambda, \quad \alpha$

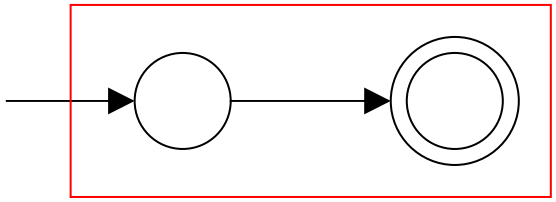
AFNs



$$L(M_1) = \varnothing = L(\varnothing)$$



$$L(M_2) = \{\lambda\} = L(\lambda)$$



$$L(M_3) = \{a\} = L(a)$$

Linguagens Regulares

Hipotese Indutiva

Assuma

Para ERs r_1 e r_2

que

$L(r_1)$ e $L(r_2)$ são LRs

Passo indutivo

Devemos provar que:

$$\left.\begin{array}{l} L(r_1 + r_2) \\ \\ L(r_1 \cdot r_2) \\ \\ L(r_1 *) \\ \\ L((r_1)) \end{array}\right\} \text{São LRs}$$

ERs por definição:

$$L(r_1 + r_2) = L(r_1) \cup L(r_2)$$

$$L(r_1 \cdot r_2) = L(r_1)\, L(r_2)$$

$$L(r_1 *) = (L(r_1))*$$

$$L((r_1)) = L(r_1)$$

Pela hipótese indutiva sabemos que:

$L(r_1)$ e $L(r_2)$ são LRs

Sabemos também que:

LRs são fechadas sobre

união	$L(r_1) \cup L(r_2)$
concatenação	$L(r_1)\, L(r_2)$
estrela	$(L(r_1))^*$

Então:

$$L(r_1 + r_2) = L(r_1) \cup L(r_2)$$

$$L(r_1 \cdot r_2) = L(r_1)\, L(r_2)$$

$$L(r_1 *) = (L(r_1))*$$

São LRs

E trivialmente:

$$L((r_1)) \quad \text{É}$$

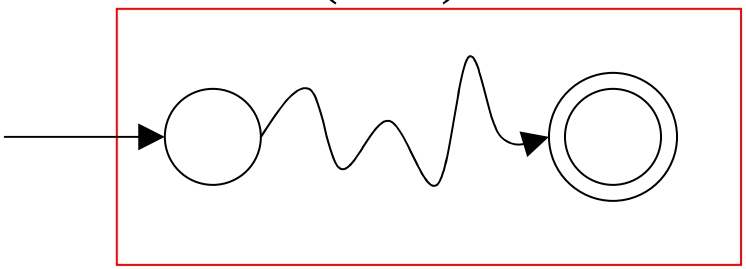
Prova - Parte 2

2. Para qualquer LR L existe uma ER r com $L(r) = L$

Porva por construção da ER

Desde que L é regular tomamos um AFN M que aceita L

$$L(M) = L$$



Único estado final

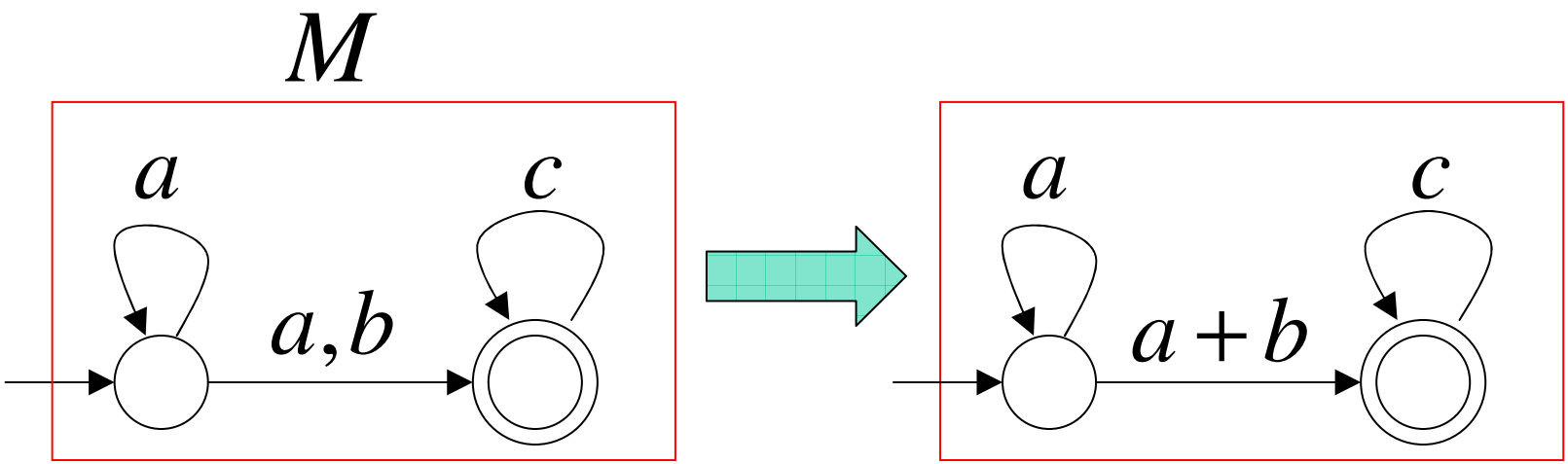
De M construímos um

Grafo de transição generalizado

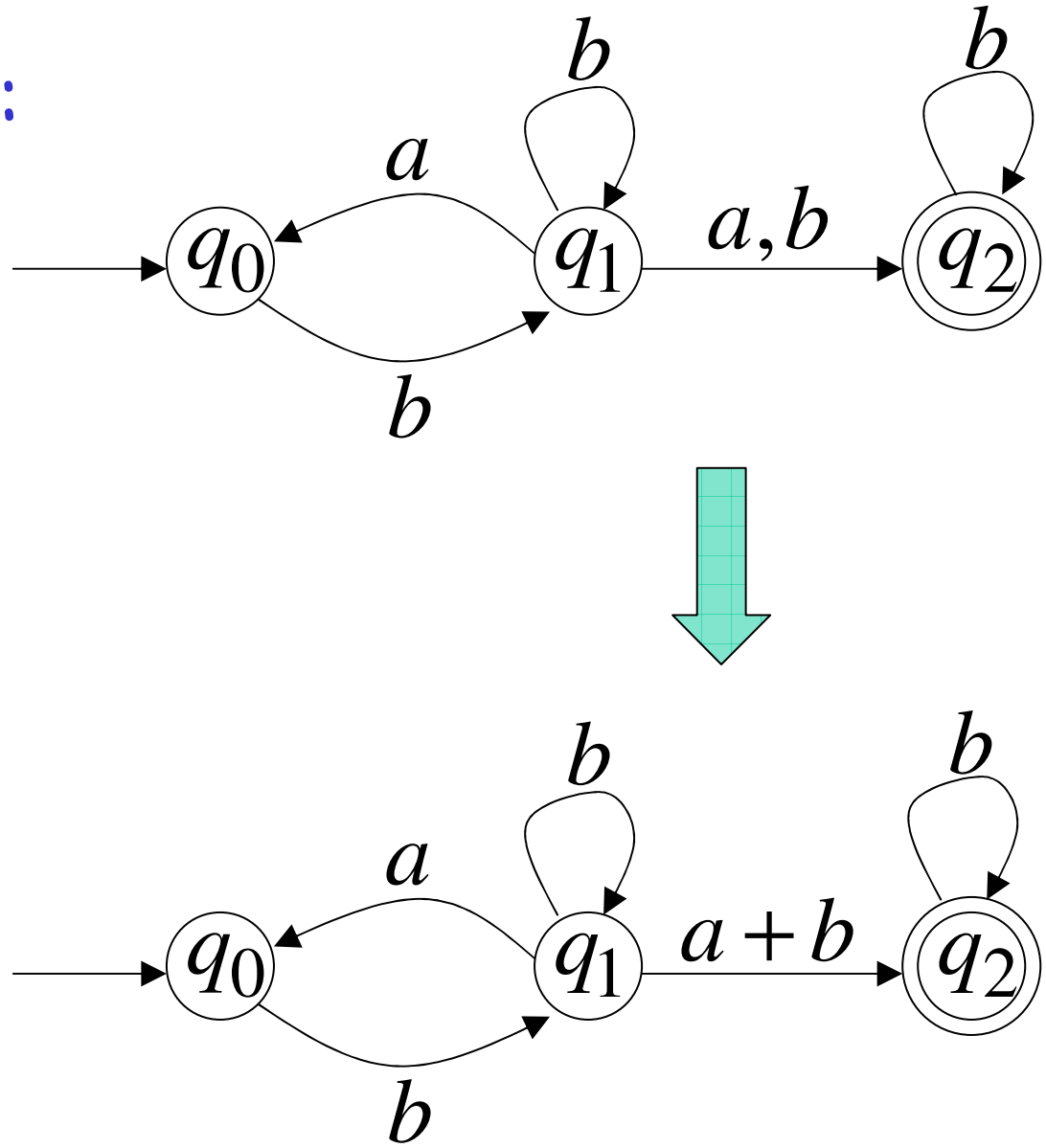
rótulos das transições

são expressões regulares

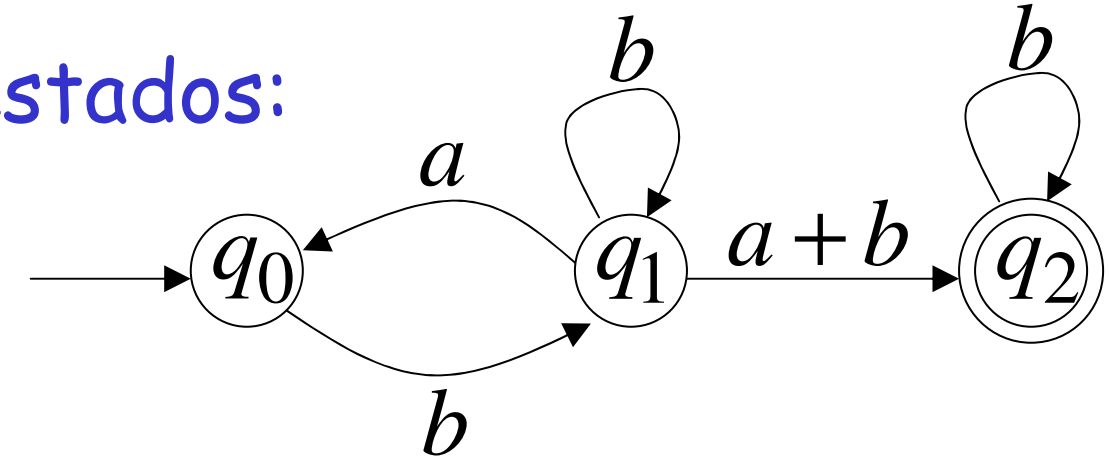
Exemplo:

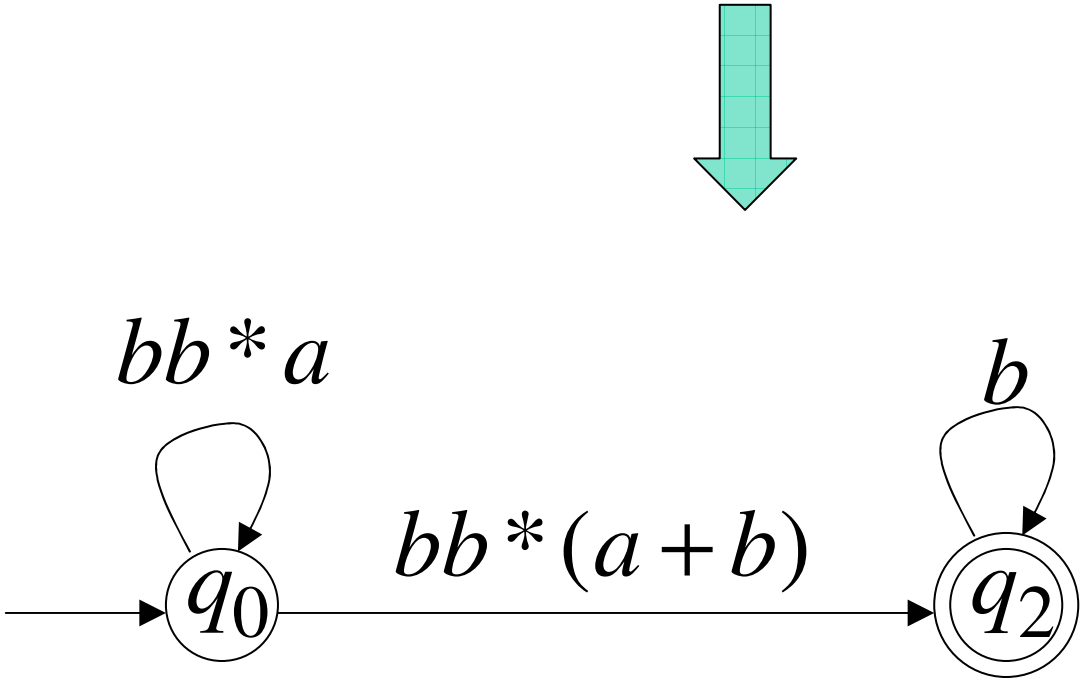


Outro exemplo:

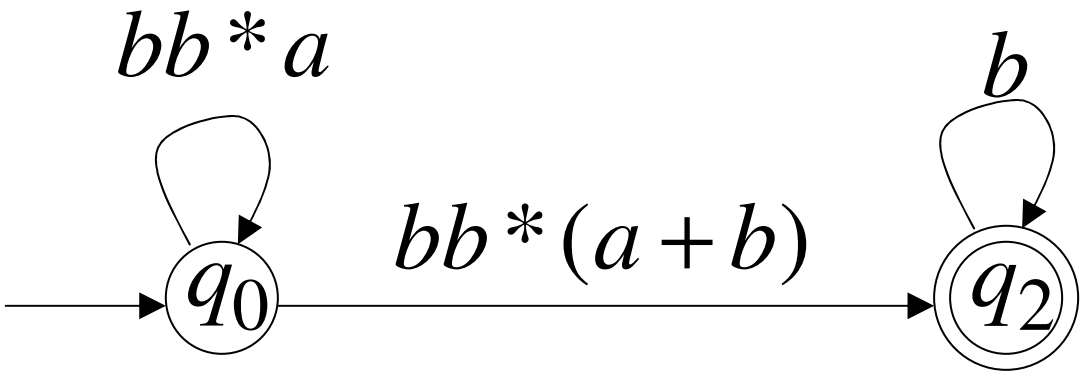


Reduzindo os estados:





Resultando na Expressão Regular:

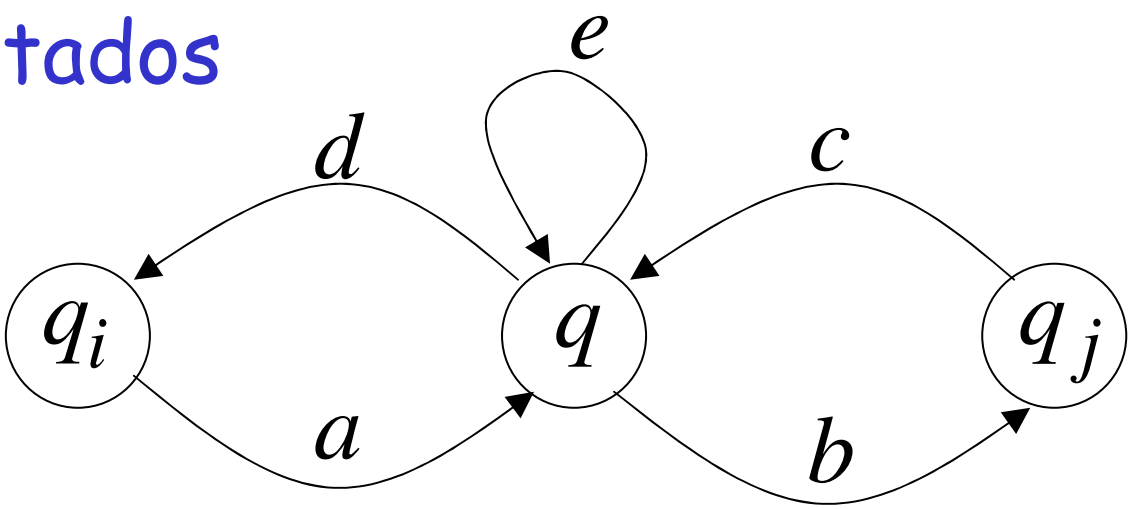


$$r = (bb^*a)^*bb^*(a+b)b^*$$

$$L(r) = L(M) = L$$

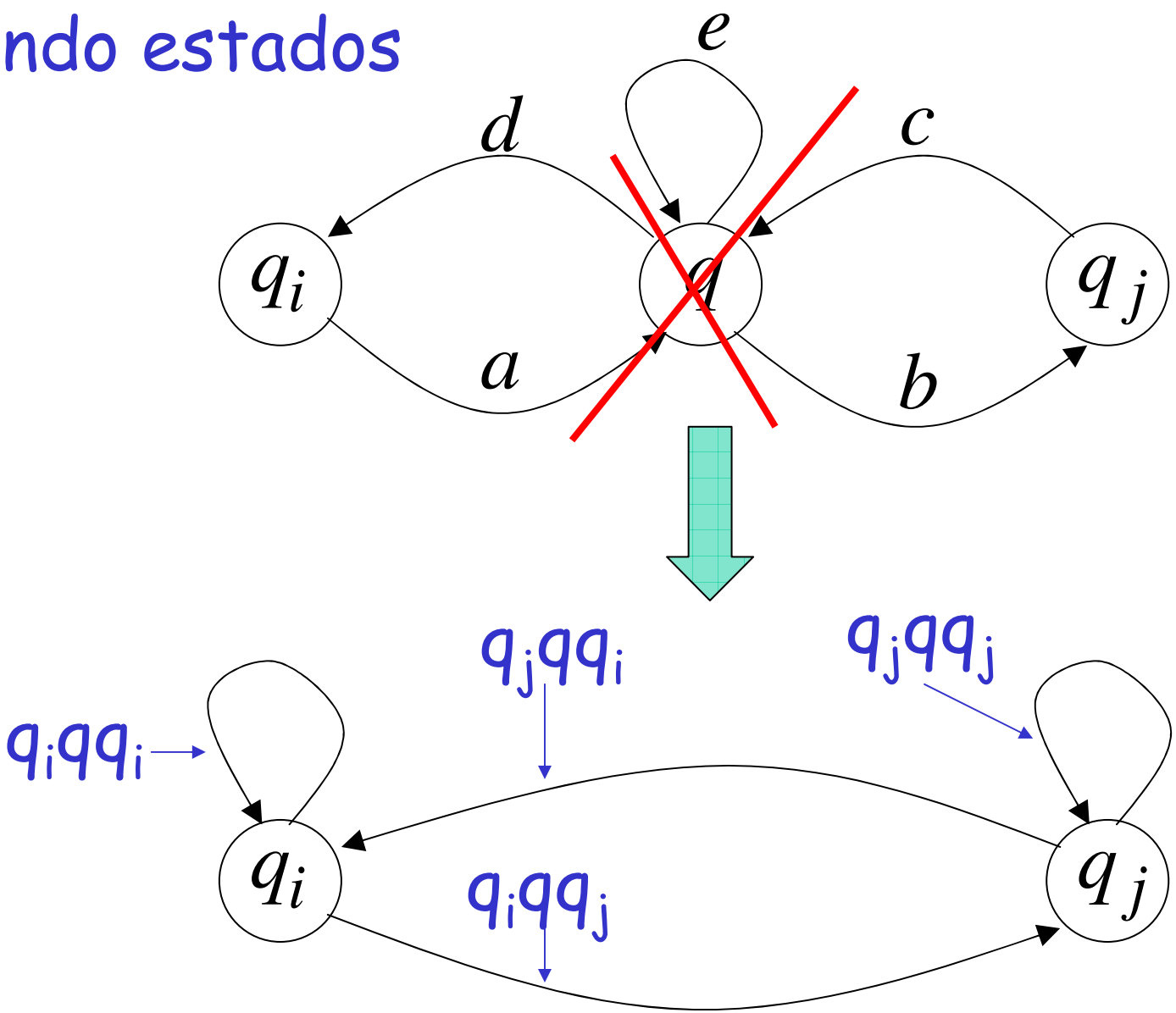
Em geral

Removendo estados



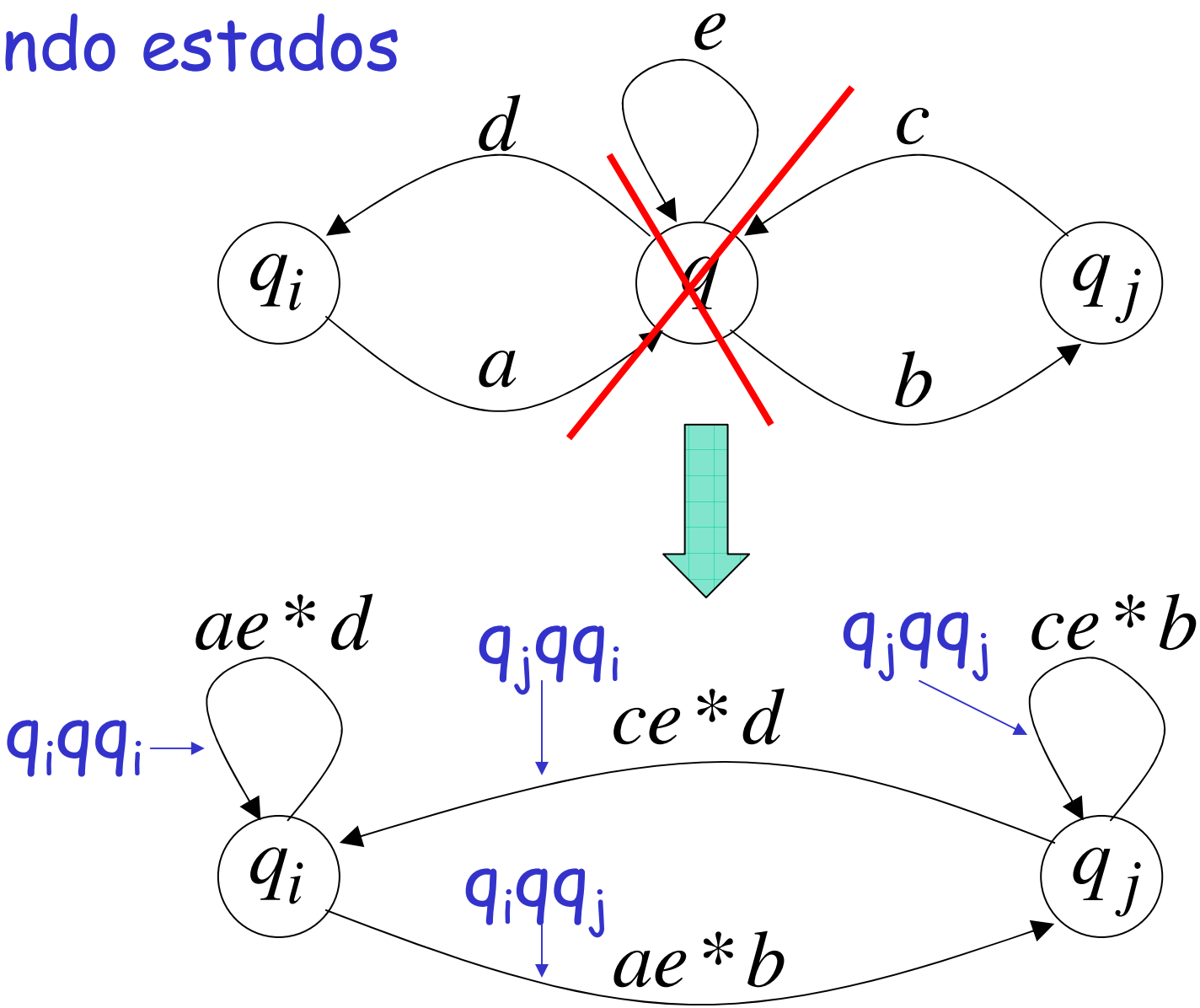
Em geral

Removendo estados

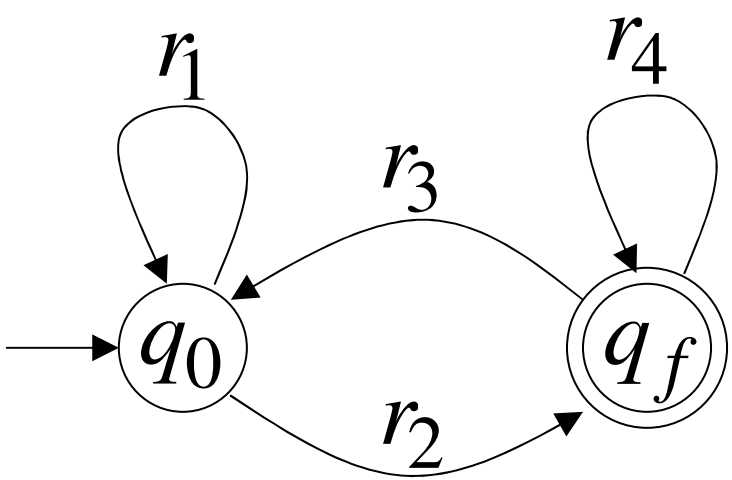


Em geral

Removendo estados



Obtendo uma expressão regular final



$$r = r_1 * r_2 (r_4 + r_3 r_1 * r_2) *$$

$$L(r) = L(M) = L$$